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30 November 1984
SUPERSEDING
MIL-T-50334A (MU)
13 February 1976

MILITARY SPECIFICATION

TEST SET, SUBASSEMBLY MX-8049/TPM-22A (RANGE COMPUTER TESTER)

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Test Set, Subassembly MX-8049/TPM-22A hereinafter referred to as range computer tester, which is a part of the Radar Test Set AN/TPM-22A. The range computer tester tests both the search and track modes of the Range Computer CP-888/VPS-2, a part of the Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-C-39012	-	Connector, Coaxial Radio Frequency, General Specification for

FSC 4931

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center Attn: DRSMC-QA Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-46357	-	Computer, Gun Direction; M18 Manufacturing Process and Inspection of
MIL-C-48581	-	Circuit Card Assembly, Target Coincidence Board, 11747210
MIL-C-48582	-	Circuit Card Assembly, Target Position Board, 11747198
MIL-C-48583	-	Circuit Card Assembly, Target Acceleration Board, 11747207
MIL-P-55149	-	Posts, Binding, Electrical

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods
MS3126	-	Connectors, Plug, Electrical, Crimp

2.1.2 Other government documents, drawings, and publications. The following other government drawings form a part of this specification to the extent specified herein.

DRAWINGS (see 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9333061	-	Test Set, Subass'y MX-8049/TPM-22A
9333076	-	Test Set, Subassembly MX-8049/TPM-22A Schem
9352607	-	Transformer, Pulse
10559851	-	Diode

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11747198	-	Printed Wiring Board Assembly (A3) MIL-C-48582 (Target Position Board)
11747207	-	Printed Wiring Board Assembly (A4) MIL-C-48583 (Target Acceleration Board)
11747210	-	Printed Wiring Board Assembly (A1) MIL-C-48581 (Target Coincidence Board)
11748023	-	Integrated Circuit, 75 MHZ and Input J-K Flip Flop

PACKAGING DATA SHEET

9333061	-	Packaging of Test Set, Subassembly MX-8049/TPM-22A

(Copies of specification, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fabrication. The range computer tester shall be manufactured in accordance with Drawing 9333061 and drawings pertaining thereto. The following subassembly shall have been inspected and accepted in accordance with the specification prior to assembly in the range computer tester.

Unit No.	Name	Specification
A1	Target Coincidence Board: 11747210	MIL-C-48581
A3	Target Position Board: 11747198	MIL-C-48582
A4	Target Acceleration Board: 11747207	MIL-C-48583

3.2 General specification. Where applicable, the range computer tester shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

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3.2.1 MIL-F-13926

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishings

d. Part identification and marking

e. Workmanship

f. Soldering electrical connections

3.2.2 MIL-C-46357

a. Etched circuit assemblies

b. Wiring

c. Threaded fasteners

d. Marking

e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. The range computer tester shall be mounted on the vibration table in normal service position. The range computer tester shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range (Hz ± 2 percent)	Excursion (double amplitude) (inches ± 10 percent
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

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3.3.2 Shock. The range computer tester shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds. The G value of each impulse shall be within ± 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds.

3.3.3 Altitude.

3.3.3.1 Operating. The range computer tester shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Non-operating. The range computer tester shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The range computer tester shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage). The range computer tester shall not suffer permanent damage when subjected to the temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 48 hours (after temperature stabilization), and shall meet the requirements 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The range computer tester shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be continuous electrically as specified on the schematic diagram (9333076) and as specified in Table I with A1, A3, A4 card subassemblies removed.

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TABLE I. Signal monitoring switch continuity

SWITCH POSITION		TERMINAL POINTS FROM	TO
DIGITAL VOLTMETER HIGH (S7)	0	J4-E*	J10
	1	J1-A	J10
	2	J1-H	J10
	3	J1-F	J10
	4	J1-C	J10
	5	J2-H	J10
	6	J2-K	J10
	7	J2-G	J10
	8	J2-A	J10
	9	J2-J	J10
	10	J3-A	J10
	11	J3-C	J10
	12	J3-B	J10
	13	J3-J	J10
	14	J3-H	J10
DIGITAL VOLTMETER RETURN (S8)	1	J1-B	J11
	2	J1-G	J11
	3	J1-G	J11
	4	J1-K	J11
	5	J2-F	J11
	6	J2-F	J11
	7	J2-F	J11
	8	J2-B	J11
	9	J2-B	J11
	10	J3-L	J11
	11	J3-L	J11
	12	J3-L	J11
	13	J3-K	J11
	14	J3-K	J11
	15	No Connection	

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TABLE I. Signal monitoring switch continuity - continued

SWITCH POSITION		TERMINAL POINTS FROM	TO
OSCILLOSCOPE CHAN A (S5)	0	J4-B*	J6*
	0	J4-B**	J6**
	1	J4-C*	J6*
	1	J4-C**	J6**
	2	J4-D*	J6*
	2	J4-D**	J6**
	3	J4-E*	J6*
	3	J4-E**	J6**
	4	J4-F*	J6*
	4	J4-F**	J6**
	5	J4-G*	J6*
	5	J4-G**	J6**
	6	J3-A	J6*
	6	J3-L	J6**
	7	J3-J	J6*
	7	J3-K	J6**
	8	J5-E	J6*
	8	J5-F	J6**
	9	J25-16	J6*
	9	J25-14	J6**
OCILLOSCOPE CHAN B (S6)	0	J4-B*	J7*
	0	J4-B**	J7**
	1	J4-C*	J7*
	1	J4-C**	J7**
	2	J4-D*	J7*
	2	J4-D**	J7**
	3	J4-E*	J7*
	3	J4-E**	J7**
	4	J4-F*	J7*
	4	J4-F**	J7**
	5	J4-G*	J7*
	5	J4-G**	J7**
	6	J3-A	J7*
	6	J3-L	J7**
	7	J3-J	J7*
	7	J3-K	J7**
	8	J5-E	J7*
	8	J5-F	J7**
	9	J25-16	J7*
	9	J25-14	J7**

* - Coaxial Pin Conductor
** - Coaxial Pin Shield

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3.5 Performance.

3.5.1 POWER switch S1.

3.5.1.1 OFF. OFF legend of S1 shall be illuminated when S1 is in normally closed position and 24 Vdc ± 5% is applied to J8-R(+) and J8-P(Ret).

3.5.1.2 ON. ON legend of S1 shall be illuminated and 24 Vdc ± 5% shall be available between J5-D(+), J5-B(+), J5-T(+), and J8-P(Ret), and between J12-A(+) and J12-B(Ret), under the following conditions:

a. S1 in normally open position.

b. 24 Vdc ± 5% applied to J8-R(+), J8-N(+), and J8-P(Ret).

c. 24V circuit breaker CB1 closed (white band not visible).

3.5.2 24V circuit breaker CB1. The 24V circuit breaker CB1 shall be closed when reset pushbutton is depressed so white band is not exposed. The circuit breaker shall be open when reset pushbutton is pulled out to expose white band.

3.5.3 POWER indicator DS3 legends.

3.5.3.1 +33V legend. The +33V legend of DS3 shall be illuminated under the following conditions:

a. Controls and conditions met in accordance with 3.5.1.2.

b. 33 Vdc ± 3% applied to J8-L(+) and J8-M(Ret).

3.5.3.2 -33V legend. The -33V legend of DS3 shall be illuminated under the following conditions:

a. Controls and conditions met in accordance with 3.5.1.2.

b. 33 Vdc ± 3% applied to J8-F(-) and J8-G(Ret).

3.5.3.3 +23V legend. The +23V legend of DS3 shall be illuminated when controls and conditions are met in accordance with 3.5.1.2 and 23 Vdc ± 3% is applied to J8-C(+) and J8-D(Ret).

3.5.3.4 -23V legend. The -23V legend of DS3 shall be illuminated when controls and conditions are met in accordance with 3.5.1.2 and 23 Vdc ± 3% is applied to J8-E(-) and J8-D(Ret).

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3.5.4 RADIATE switch S2.

3.5.4.1 OFF legend. The OFF legend of S2 shall be illuminated under conditions stated in 3.5.1.2 and S2 in normally closed position.

3.5.4.2 ON legend. The ON legend of S2 shall be illuminated and 24 Vdc ± 5% shall be available between J5-J(+) and J5-K(Ret) under conditions stated in 3.5.1.2 and S2 in normally open position.

3.5.5 TARGET/NO TARGET switch S3.

3.5.5.1 TARGET legend. With power on as stated in 3.5.1.2 set MODE switch to 400M, push STOP switch (S11), and then activate target switch S3. The S3 legend shall switch from NO TARGET to TARGET.

3.5.5.2 NO TARGET legend. With conditions established by 3.5.5.1 activate STOP switch (S11). The S3 legend shall switch from TARGET to NO TARGET.

3.5.6 SEARCH/TRACK indicator DS2.

3.5.6.1 TRACK legend. The TRACK legend of DS2 shall be illuminated when 24 Vdc ± 5% is applied to J5-W(+) and J8-P(Ret).

3.5.6.2 SEARCH legend. The SEARCH legend of DS2 shall be illuminated when 24 Vdc ± 5% is applied to J5-C(+) and J8-P(Ret).

3.5.7 DOPPLER indicator DS1.

3.5.7.1 NO ECM legend. The NO ECM legend of DS1 shall be illuminated when 24 Vdc ± 5% is applied to J5-A(+) and J8-P(Ret).

3.5.7.2 ECM legend. The ECM legend of DS1 shall be illuminated when 24 Vdc ± 5% is applied to J5-V(+) and J8-P(Ret).

3.5.8 TARGET REJECT switch S4.

3.5.8.1 The TARGET REJECT legend of S4 shall be illuminated white when conditions are as stated in 3.5.1.2 and S4 is in the normal position.

3.5.8.2 The TARGET REJECT legend of S4 shall be illuminated red and 24 Vdc ± 5% shall be available between J5-G(+) and J5-H(Ret), when conditions are as stated in 3.5.1.2 and S4 is held in depressed position.

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3.5.9 60 MHz oscillator. Under conditions stated in 3.5.1.2 and 28 Vdc ± 1% applied to J8-H(+) and J8-J(Ret), the signal at J4-A shall be 59.936 MHz ± 1000 Hz sine wave at 2.4 to 5.0 Vac peak-to-peak into a 100 ohm load. The reading at J4-A shall be zero with POWER switch S1 in the OFF position.

3.5.10 120 MHz clock pulse.

3.5.10.1 120 MHz signal. Under conditions stated in 3.5.1.1, 28 Vdc ± 1% applied to J8-H(+) and J8-J(Ret), 24 Vdc ± 5% applied to J8-N(+) and J8-P(Ret), and TEST MODE switch S10 momentarily actuated, the signal at J28 shall be 119.870 MHz ± 200 Hz sine wave, at 700 MV peak-to-peak min. into a 50 ohm load.

3.5.10.2 No 120 MHz signal. Under conditions stated in 3.5.1.2, and 28 Vdc ± 1% applied to J8-H(+) and J8-J(Ret), there shall be 0 ± .05 Vdc present at J28.

3.5.11 Target Trigger. With OSCILLOSCOPE CHAN A switch (S5) set to position 9, the Target Trigger pulse shall be present at OSCILLOSCOPE CHAN A Terminal (J6) and shall be a positive pulse from 0 V to not less than 15 Vdc. The pulse width shall be between 0.2 and 0.3 usec when the pulse at TP1 of Target Coincidence board (A1) is adjusted to 0.35 ± 0.05 usec.

3.5.12 Audio. Under conditions stated in 3.5.1.2, 28 Vdc ± 1% applied to J8-H(+) and J8-J(Ret), and input signal Item 1 of Table II, the signal at AUDIO terminal (J29) shall be a 1 KHz ± 10 Hz Square Wave, 0 V to 4.0V $^{+\ 1.0}_{-\ 1.5}$ V.

TABLE II. Signal inputs

ITEM	SIGNAL	CHARACTERISTICS	APPLICATION POINTS
1	Mod to Trigger	Positive Pulse from 0 V to 22 ± 4 Vdc. 0.2 to 0.6 usec Pulse Width @ 50% points. Rise time of 40 nsec max into 50 ohms as measured between 10% and 90% points. 20 KHz ± 200 Hz PRF.	J4-E
2	E/L Trigger	Positive Pulse from 0 V to 4.0 ± 0.5 Vdc. Pulse width manually variable between 5 usec and 10 usec. Falling edge occurring at rising edge of Mod To. PRF = 20 KHz ± 200 Hz.	J4-D

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3.5.13 Simulated target signals.

3.5.13.1 Simulated target. The simulated target pulse shall be present at OSCILLOSCOPE CHAN A (J6), position 9, and shall be a positive pulse from 0 V to not less than 3 Vdc, with a pulse width between 0.1 usec and 0.3 usec, and a PRF = 20 KHz ± 200 Hz. It shall have the characteristics described in Table III when the following conditions are applied:

a. Controls and conditions met in accordance with 3.5.3.2.

b. 28V ± 0.2 Vdc applied to J8-H(+), and J8-J(Ret).

c. Input signal Item 1 applied as specified in Table II.

d. MODE switch S9 set as stated in 3.5.13.2.

e. DIRECTION switch S12 set as stated in 3.5.13.3.

f. TARGET/NO TARGET switch S3 depressed as stated in 3.5.13.4.

g. STOP switch S11 depressed as stated in 3.5.13.5.

3.5.13.2 Mode switch S9. The position of the MODE switch shall determine the speed and acceleration of the simulated target described in 3.5.13.1. The target's performance when affected by this switch is tabulated in Table III.

3.5.13.3 DIRECTION switch S12. The position of the DIRECTION switch shall determine the direction of the 70M/S and 240M/S targets only as selected on the MODE switch (3.5.13.2). No other targets shall be affected. The target's performance when affected by this switch is tabulated in Table III.

3.5.13.4 TARGET/NO TARGET switch S3. The TARGET/NO TARGET switch is a spring loaded momentary actuated switch. When the NO TARGET indicator (DS6) is illuminated and the TARGET indicator is extinguished there shall be no simulated target pulse present at OSCILLOSCOPE CHAN A terminal (J6). When the TARGET/NO TARGET switch is depressed, the NO TARGET indicator shall extinguish, the TARGET indicator shall illuminate and the simulated target pulse shall be present at OSCILLOSCOPE CHAN A terminal (J6). When the simulated target has finished its run as described in Table III, the NO TARGET indicator shall illuminate and the TARGET indicator shall extinguish.

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TABLE III. Simulated target signals.

RUN	TYPE	MODE SWITCH	DIRECTION SWITCH	DESCRIPTION
1	400m, stationary	400m	Note 1	Stationary pulse at 7.5 to 8.5 usec after leading edge of T_o.
2	4000m, stationary	4000m	Note 1	Note 2. Stationary pulse at 31 to 33 usec after leading edge of MOD Trigger.
3	70m/s, outbound	70m/s	OUT	Pulse moves along the target path in 73 to 78 seconds. The pulse shall exist between 34 and 36 usec of PRR with a 1 sec fade occuring between 16 to 17 usec. Note 3.
4	70m/s, inbound	70m/s	IN	Pulse moves along the target path in 73 to 77 seconds. The pulse shall exist between 34 and 36 usec of PRR with a 1 sec fade occuring between 16 to 17 usec. Note 3.
5	240m/s, outbound	240m/s	OUT	Pulse moves from T_o + 1.0 usec to T_o +34 ± 1 usec in 20 to 24 sec. Note 3.
6	240m/s, inbound	240m/s	IN	Pulse moves from T_o + 34 ± 1 usec to T_o in 20 to 24 sec. Note 3.
7	20m/s^2, accelerating	20m/s^2	Note 1	Pulse moves from T_o + 34 to 36 usec to T_o + 1.0 usec and back then out to T_o + 34 ± 1 usec in 64 to 67 sec.
8	80m/s^2, accelerating	80m/s^2	Note 1	Pulse moves in from T_o to 34 ± 1 usec to T_o +27 ± 2 usec and then out to T_o +34 ± 1 usec in 16 to 18 sec.

Note 1: Either position.
Note 2: This run shall be present until manually terminated by STOP switch.
Note 3: Target shall fade for 1 ± 0.2 sec duration at position T_o to + 17 ± 1 usec.

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3.5.13.5 STOP switch S11. Given that a simulated target is generated per 3.5.13.1, the momentary depression of the STOP switch shall at any time terminate the target and cause the NO TARGET indicator to illuminate and the TARGET indicator to extinguish.

3.5.13.6 Target fade. Given that a simulated target is generated per 3.5.13.1, the target will disappear for a duration of 1 ± 0.2 SEC at position To + 17 ± 1 usec, for those noted in Table III. During this fade condition the TARGET indicator (DS6) shall remain illuminated and the NO TARGET indicator shall remain extinguished.

3.5.14 Doppler signals.

3.5.14.1 Sum doppler. The sum doppler signal shall be present at J4-F and have the characteristics described in Item 1 of Table IV under the following conditions:

a. Controls and conditions met in accordance with 3.5.13.1, Run 2.

b. SUM level control R2 fully clockwise.

c. 150 ohm resistive load at J4-F.

d. Input signal Item 2 applied as specified in Table II.

e. The pulse at TP3 on Target Coincidence board (A1) is coincident with either the pulse at TP1 or the pulse at TP2 on Target Coincidence board (A1). When there is no coincidence, there shall be a dc level of -.86 ± .04 Vdc.

3.5.14.2 Difference doppler (In Phase). The difference doppler signal shall be present at J4-G and have the characteristics described in Item 2 of Table IV under the following conditions:

a. Controls and conditions met in accordance with 3.5.13.1, Run 2.

b. DIFFERENCE level control R3 fully clockwise.

c. 150 ohm resistive load at J4-G.

d. Input signal Item 2 applied as specified in Table II.

e. The pulse at TP3 on Target Coincidence board (A1) is at least 25 ± 2 nsec coincident with the pulse at TP1 on Target Coincidence board (A1). When there is no coincidence, there shall be a dc level of -.86 ± .04 Vdc.

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3.5.14.3 Difference doppler (Out of Phase). The difference doppler signal shall be present at J4-G and have the characteristics described in Item 3 of Table IV under the following conditions:

a. Controls and conditions met in accordance with 3.5.13.1, Run 2.

b. DIFFERENCE level control R3 fully clockwise.

c. 150 ohm resistive load at J4-G.

d. Input signal Item 2 applied as specified in Table II.

e. The pulse at TP3 on Target Coincidence board (A1) is at least 25 ± 2 nsec coincident with the pulse at TP2 on Target Coincidence board (A1). When there is no coincidence, there shall be a dc level of -.86 ± .04 Vdc.

3.6 First Article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.7 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

TABLE IV. Sum and difference signals.

ITEM	SIGNAL OUTPUT	CHARACTERISTICS	APPLICATION POINTS
1	Sum Doppler	1 KHz ± 100 Hz square wave, from 0 V to not less than 1.5V (peak-to-peak). Amplitude controlled by SUM level control R2.	J4-F
2	In Phase Difference Doppler	1 KHz ± 100 Hz square wave, from 0 V to not less than 1.5V (peak-to-peak), in phase with sum doppler. Amplitude controlled by DIFFERENCE level control R3.	J4-G
3	Out of Phase Difference Doppler	1 KHz ± 100 Hz square wave, from 0 V to not less than 1.5V (peak-to-peak), out of phase with sum doppler. Amplitude controlled by DIFFERENCE level control R3.	J4-G

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Unless otherwise specified in the item detail specification, or contract, the following types of inspection shall be conducted on the unit or product:

a. First article inspection (4.3)

b. Quality conformance inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer (see 6.2) for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 1 test set.

4.3.2 Inspection to be performed. First article sample may be subjected by the Government to any or all of the examinations and tests specified in 4.4.2 of this specification and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revision and complies with the provisions for submission of product in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.4.2 Examinations and tests. Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables V and VI and methods of inspection in 4.7. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. Workmanship shall be in accordance with MIL-STD-454 and other applicable specifications.

TABLE V. Classification of defects.

100% Inspection

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Electrical	3.4	4.7.2
102.	POWER switch S1	3.5.1	4.7.3.2
103.	24V circuit breaker CB1	3.5.2	4.7.3.2.2
104.	POWER indicator legends DS3	3.5.3	4.7.3.3
105.	RADIATE switch S2	3.5.4	4.7.3.4
106.	TARGET/NO TARGET switch S3	3.5.5	4.7.3.5
107.	SEARCH/TRACK indicator DS2	3.5.6	4.7.3.6

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TABLE V. Classification of defects - continued
100% Inspection

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
108.	DOPPLER indicator DS1	3.5.7	4.7.3.7
109.	TARGET REJECT switch S4	3.5.8	4.7.3.8
110.	60 MHz oscillator	3.5.9	4.7.3.9
111.	120 MHz clock pulse	3.5.10	4.7.3.10
112.	Target Trigger	3.5.11	4.7.3.11
113.	Audio	3.5.12	4.7.3.12
114.	Simulated target signals	3.5.13	4.7.3.13
115.	Doppler signals	3.5.14	4.7.3.14

4.4.3 Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Special sampling. One receiver test panel out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table V and shall then meet the requirements and tests in Table VI.

TABLE VI. Special sampling.

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General specification	3.2	MIL-F-13926/ MIL-C-46357 visual
303.	Vibration	3.3.1	4.7.1.1
304.	Shock	3.3.2	4.7.1.2
305.	Altitude (operating)	3.3.3.1	4.7.1.3
306.	Altitude (non-operating)	3.3.3.2	4.7.1.3
307.	Temperature	3.3.4	4.7.1.4
308.	Humidity (non-operating)	3.3.5	4.7.1.5

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and a re-submitted sample has been approved. (see 4.4.3).

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4.6 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the referenced specifications and standards cited in 4.4.2. Inspection equipment shall be as listed in Table VII. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See 6.3 of this specification.

4.6.1 Inspection equipment. The following inspection equipment or equivalent shall be used in the performance of the applicable test as specified in 4.7.

TABLE VII. Inspection equipment list.

Item Number		Description
1.	Power Supply	5.00 ± 0.25 Vdc, regulated, AC ripple less than 50 mv peak-to-peak, capacity 500 ma.
2.	Power Supply	24.0 ± 1.2 Vdc, regulated, AC ripple less than 50 mv peak-to-peak, capacity 3 ampere.
3.	Power Supply	28.0 ± 0.2 Vdc, regulated, AC ripple less than 50 mv peak-to-peak, capacity 500 ma.
4.	Power Supply	33.0 ± 0.3 Vdc, regulated, AC ripple less than 100 mv peak-to-peak, capacity 500 ma.
5.	Power Supply	0-40 Vdc, regulated, AC ripple less than 100 mv peak-to-peak, capacity 200 ma.
6.	Oscilloscope	Dual channel; rise time of scope plus probe not greater than 10 nsec, sweep period less than 50 nsec, Input Resistance: 8 megohms or greater.
7.	Test Fixture	As specified in Fig. 1. See Table VIII for test fixture parts list.
8.	Digital Voltmeter	0 to 40 Vdc, .1% accuracy, input impedance greater than 10 megohms.
9.	Ohmmeter	0 to 200 ohms, ± 1% accuracy.
10.	Frequency Counter	Sensitivity of 0.1V rms capable of counting 120 MHz ± 200 Hz.
11.	Resistor, Load	100 ohms, 1/2W ± 5%.
12.	Resistor, Load	50 ohms, 1/2W ± 5%.

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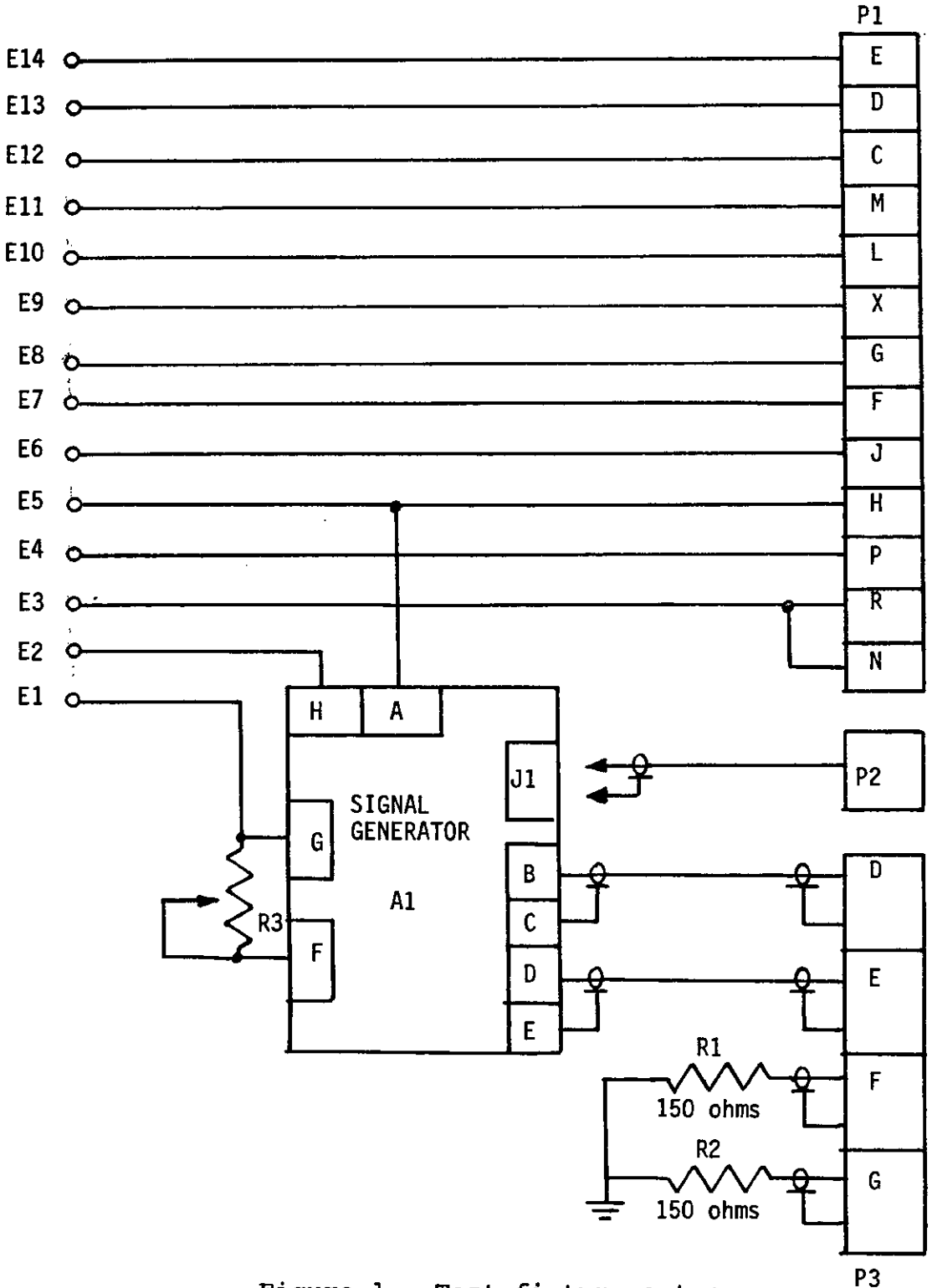


Figure 1. Test fixture setup.

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TABLE VIII. Test fixture parts list.

DESIGNATION	NOMENCLATURE AND DESCRIPTION
A1	Test fixture signal generator in accordance with Figure 2. See Table IX for parts list.
P1	Connector - P/N MS3126F22-21SZ
P2	Connector - P/N M39012/16-0001
P3	Connector - Bendix P/N PTSF06SE-22-78P
R1, R2	150 ohm, 1/4 watt, 5% resistor
R3	Variable resistor, 10K, > 1/4 watt 10 turn
E1, E3, E5, E7, E10, E12, E14	Binding post, PB01FA03 per MIL-P-55149/1
E2, E4, E6, E8, E9, E11, E13	Binding post, PB01FA04 per MIL-P-55149/1

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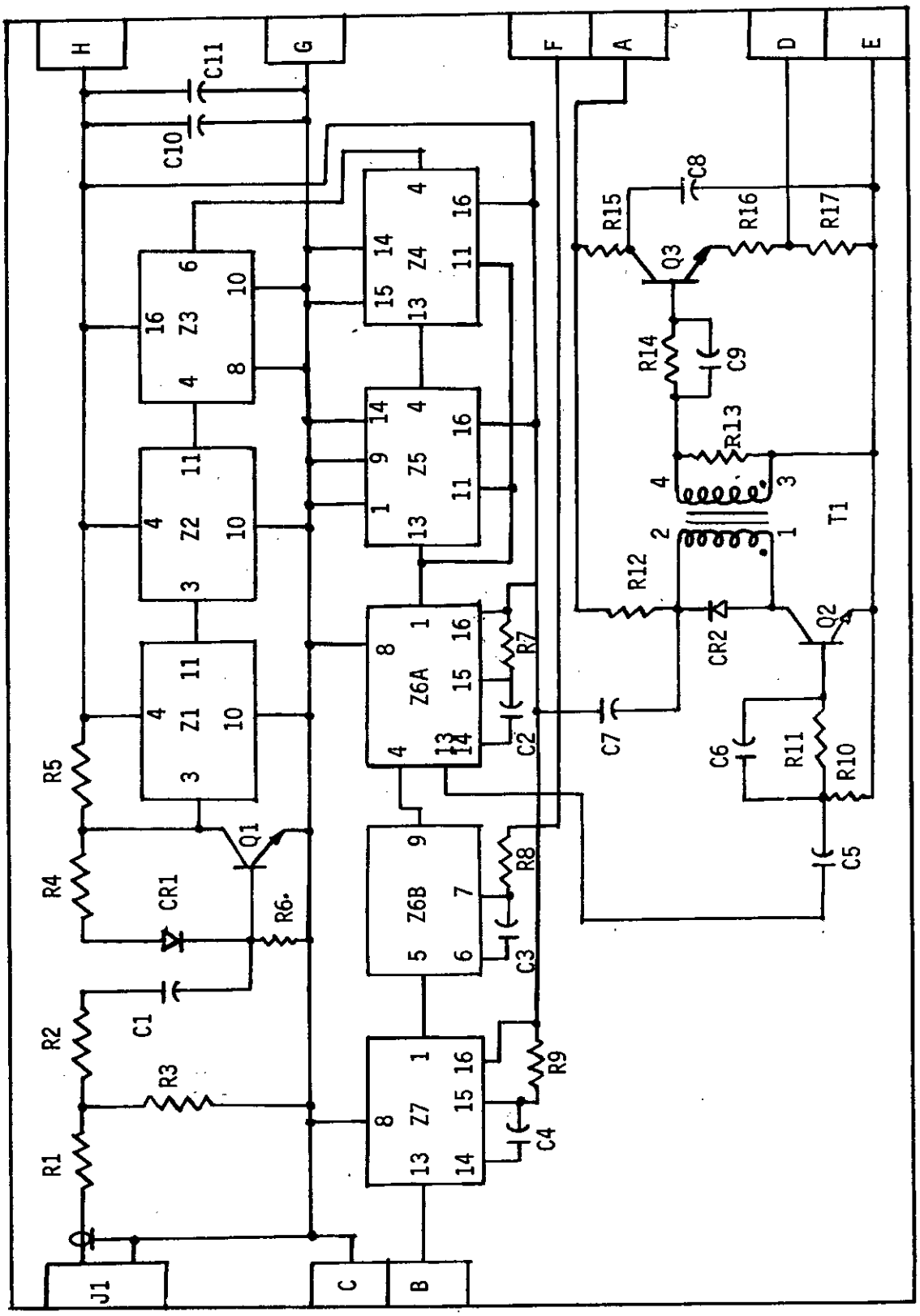


Figure 2. A1 - Test fixture signal generator.

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TABLE IX. Test fixture signal generator parts list.

DESIGNATION	NOMENCLATURE AND DESCRIPTION
Q1	Transistor, 2N706B
Q2, Q3	Transistor, 2N697
CR1	Diode, P/N 10559851
CR2	Diode, 1N914
Z1, Z2	Integrated Circuit, P/N 11748023
Z3, Z4, Z5	Integrated Circuit, Texas Instruments P/N SN54193J
Z6, Z7	Integrated Circuit, Texas Instruments P/N SN54123J
T1	Transformer, Pulse P/N 9352607-2
R1, R2	Resistor, 16 ohm ± 5%, 1/4 watt
R3	Resistor, 68 ohm ± 5%, 1/4 watt
R4, R15	Resistor, 220 ohm ± 5%, 1/4 watt
R5	Resistor, 100 ohm ± 5%, 1/4 watt
R6	Resistor, 470 ohm ± 5%, 1/4 watt
R7	Resistor, 10K ohm ± 5%, 1/4 watt
R8	Resistor, 3K ohm ± 5%, 1/4 watt
R9	Resistor, 7.5K ohm ± 5%, 1/4 watt
R10, R13, R14	Resistor, 1K ohm ± 5%, 1/4 watt
R11	Resistor, 150K ohm ± 5%, 1/4 watt
R12	Resistor, 3.3K ohm ± 5%, 1/4 watt
R16	Resistor, 15 ohm ± 5%, 1/4 watt
R17	Resistor, 270 ohm ± 5%, 1/4 watt
C1	Capacitor, .01 uF ± 10% @ 50Vdc
C2, C4	Capacitor, 200 pF ± 10% @ 50Vdc
C3	Capacitor, 4700 pF ± 10% @ 50Vdc
C5	Capacitor, 220 pF ± 10% @ 50Vdc
C6	Capacitor, 47 pF ± 10% @ 50Vdc
C7, C8, C10, C11	Capacitor, .1 uF ± 10% @ 50Vdc
C9	Capacitor, 330 pF ± 10% @ 50Vdc

4.7 Methods of inspection.

4.7.1 Environmental. Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Procedure I, Category 1, of MIL-STD-810 except as modified by paragraph 3.3.1 herein. After vibration, the range computer tester shall be tested as given in the subparagraphs of 4.7.2 and 4.7.3 for compliance with 3.3.1.

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4.7.1.2 Shock. This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be as specified in 3.3.2. After the shock impulses, the range computer tester shall be tested as specified in 4.7.2 and 4.7.3 for compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501, MIL-STD-810, with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7 the operational checkout shall be performed at the end of the tenth cycle only at standard ambient conditions.

4.7.2 Electrical. With card subassemblies A1, A3, and A4 removed, continuity shall be determined by visual examination, and an ohmmeter, item 9, Table VII for compliance to the requirements of 3.4. Subassemblies shall be replaced at the conclusion of the test.

4.7.3 Performance.

4.7.3.1 Conditions of operation. Unless otherwise specified all tests in 4.7.3 shall be performed with the D.C. power supply connections illustrated in Figure 3. The test fixture, Item 7 of Table VII, shall be connected to the applicable terminals of the Range Computer Tester as illustrated in Figure 3. Power supplies 1 through 5 of Table VII shall be connected to the designated terminals of the test fixture in compliance with the test setup of Figure 3. All power supplies shall be adjusted to the output voltage values indicated in the test setup, and turned off. The oscilloscope shall be initially disconnected.

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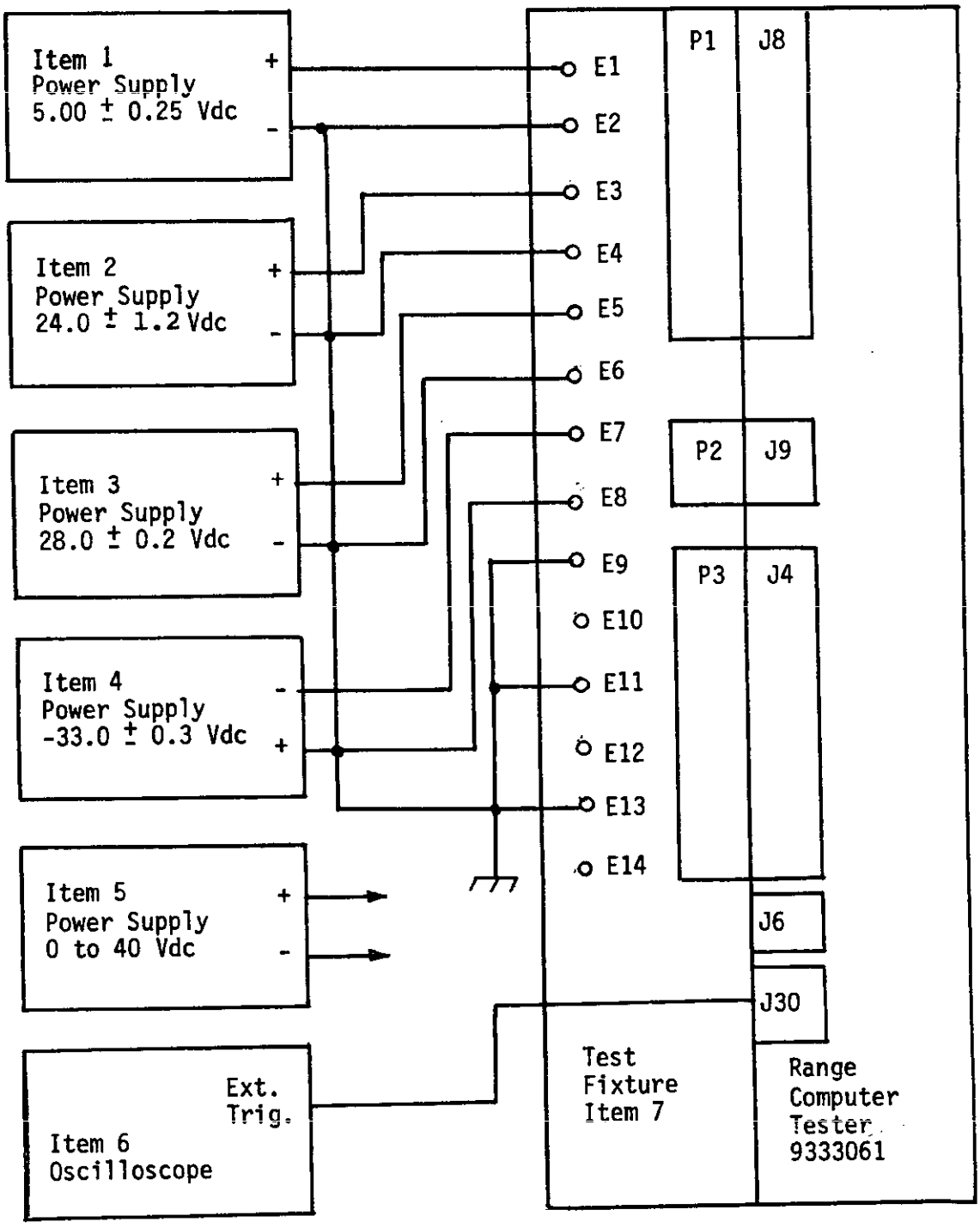


Figure 3. Range computer tester test setup.

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4.7.3.2 POWER switch S1.

4.7.3.2.1 OFF legend. Turn on the 24.0V power supply, item 2 of Figure 3. Depress and release S1 three times. Verify that OFF legend of the POWER switch is illuminated on alternate operations of S1.

4.7.3.2.2 ON legend. Set up conditions of 3.5.1.2 by turning on power supply, item 2 of Figure 3, placing S1 in the normally open position, and closing the 24V circuit breaker CB1 (white band not visible). The ON legend of S1 shall be illuminated under these conditions. Depress and release S1 three times. Verify that the OFF and ON legends of the Power Switch are alternately illuminated. Depress S1 and leave the ON legend illuminated. Use the voltmeter, item 8, of Table VII to determine the presence of a 24 Vdc ± 5% signal between J5-D(+), J5-B(+), J5-T(+), and J8-P(Ret). In like manner check for the presence of this signal between J12-A(+) and J12-B(Ret) in compliance with the requirements of 3.5.1.2. Pull out reset pushbutton of circuit breaker CB1 to expose the white band. Check to see that the circuit breaker has opened per the requirements of 3.5.2 by determining that the 24V signal is no longer present at terminals J12-A(+) and J12-B(Ret).

4.7.3.3 Power indicator DS3 legends.

4.7.3.3.1 +33V legend. Connect the variable power supply, item 5 of Figure 3, to E10(+) and E11(-) on the test fixture. Adjust the power supply to a voltage of 33 Vdc ± 3% using the digital voltmeter, item 8 of Table VII to determine the voltage. Set up conditions as stated in 3.5.1.2. Verify that +33V legend of DS3 is illuminated. Turn off and disconnect the variable power supply.

4.7.3.3.2 -33V legend. Set up conditions as stated in 3.5.1.2. Turn on -33V power supply, item 4 of Figure 3. Verify that -33V legend of DS3 is illuminated per the requirements of 3.5.3.2. Turn off the power supply.

4.7.3.3.3 +23V legend. Set up conditions as stated in 3.5.1.2. Connect the variable power supply, item 5 of Figure 3, to E12(+) and E13(-) on the test fixture. Adjust the power supply to a voltage of 23 Vdc ± 3%. Verify that +23V legend of DS3 is illuminated. Turn off and disconnect the variable power supply.

4.7.3.3.4 -23V legend. Set up conditions as stated in 3.5.1.2. Connect the variable power supply, item 5 of Figure 3, to E14(-) and E13(+) on the test fixture. Adjust the power supply to a voltage of 23 Vdc ±3%. Verify that -23V legend of DS3 is illuminated. Turn off and disconnect the variable power supply.

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4.7.3.4 RADIATE switch S2. Set up conditions as stated in 3.5.1.2. Verify that OFF or ON legend of RADIATE switch S2 is illuminated. Depress and release RADIATE switch S2 three times. Verify that OFF and ON legends are illuminated alternately with operations of S2. Verify with digital voltmeter, item 8 of Table VII, that 24 Vdc ± 5% is present between J5-J(+) and J5-K(Ret) when RADIATE ON legend is illuminated.

4.7.3.5 TARGET/NO TARGET switch S3.

4.7.3.5.1 TARGET legend. Set up conditions as stated in 3.5.1.2. Set MODE switch (S9) to 400M. Depress and release the STOP switch (S11). Depress and release the RADIATE switch (S3). Verify that the TARGET legend of S3 illuminates and stays illuminated, and that the NO TARGET legend is extinguished.

4.7.3.5.2 NO TARGET legend. Set up conditions as stated in 4.7.3.5.1. Depress and release the STOP switch (S11). Verify that the NO TARGET legend is illuminated and that the TARGET legend is extinguished.

4.7.3.6 SEARCH/TRACK indicator DS2.

4.7.3.6.1 TRACK legend. Adjust variable power supply, item 5 of Figure 3, to 24 Vdc ± 5%, and apply to terminals J5-W(+) on range computer tester and E4(-) on test fixture. Verify that TRACK legend of DS2 is illuminated. Remove power supply leads.

4.7.3.6.2 SEARCH legend. Adjust variable power supply, item 5 of Figure 3, to 24 Vdc ± 5%, and apply to terminals J5-C(+) on range computer tester and E4(-) on test fixture. Verify that SEARCH legend of DS2 is illuminated. Remove power supply leads.

4.7.3.7 DOPPLER indicator DS1.

4.7.3.7.1 NO-ECM legend. Adjust variable power supply, item 5 of Figure 3, to 24V ± 5%, and apply to terminals J5-A(+) on range computer tester and E4(-) on test fixture. Verify that NO ECM legend of LS1 is illuminated. Remove power supply leads.

4.7.3.7.2 ECM legend. Adjust variable power supply, item 5 of Figure 3, to 24V ± 5%, and apply to terminals J5-V(+) on range computer tester and E4(-) on test fixture. Verify that ECM legend of DS1 is illuminated. Remove power supply leads.

4.7.3.8 TARGET REJECT switch S4. Set up conditions as stated in 3.5.1.2. Verify that the TARGET REJECT legend is illuminated white. Depress and release S4 three times. Verify that TARGET REJECT is illuminated red while S4 is depressed and returns to white when S4 is released. Verify with digital voltmeter, item 8 of Table VII, that 24 Vdc ± 5% is present between J5-G(+) and J5-H(Ret) while S4 is depressed.

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4.7.3.9 60 MHz oscillator. Set up conditions stated in 3.5.1.2. Turn on 28V power supply, item 3 of Figure 3. Connect frequency counter, item 10 of Table VII; and oscilloscope, item 6 of Table VII, to J4-A with 50 ohm coax. Terminate coax at oscilloscope with a 100 ohm load, item 11 of Table VII. Verify that the signal present at J4-A is a sine-wave with a frequency of 59.936 MHz ± 1000 Hz, at 2.4 to 5.0 Vac peak-to-peak amplitude. Place POWER switch S1 in the OFF position and verify that the output at J4-A goes to zero.

4.7.3.10 120 MHz clock pulse.

4.7.3.10.1 120 MHz signal. Set POWER switch S1 to the OFF position, turn on 24V power supply, item 2 of Figure 3, and turn on 28V power supply, item 3 of Figure 3. Connect frequency counter, item 10 of Table VII; and oscilloscope, item 6 of Table VII, to J28 with 50 ohm coax. Terminate coax at oscilloscope with a 50 ohm load, item 12 of Table VII. Momentarily depress TEST MODE switch S10. Verify that the signal present at J28 is a 119.870 MHz ± 2000 Hz sine wave, with a peak-to-peak amplitude of 700 MV minimum. Place S1 in the ON position.

4.7.3.10.2 NO 120 MHz signal. Set up conditions as stated in 3.5.1.2, turn on 28V power supply, item 3 of Figure 3; and connect the oscilloscope, item 6 of Table VII, to J28 with 50 ohm coax. Terminate coax at oscilloscope with a 50 ohm load, item 12 of Table VII. Verify that no signal is present at terminal J28 per the requirements of 3.5.10.2.

4.7.3.11 Target Trigger. The directions given below shall be followed in sequence.

a. Attach oscilloscope, item 6 of Table VII, to OSCILLOSCOPE CHAN A, terminal (J6). Set the vertical sensitivity to 5V/CM and the horizontal rate to 1 uSEC/CM.

b. Set up conditions as specified in 3.5.1.2.

c. Turn on -33V power supply, item 4 of Figure 3.

d. Turn on 28V power supply, item 3 of Figure 3.

e. Confirm the presence of input signal, item 1 of Table II, at terminal J4-E.

f. Verify that the pulse width at TP1 on Target Coincidence board (A1) is 0.35 ± 0.02 usec at 50% points. Adjust R4 located on the Target Coincidence board if necessary.

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g. Set OSCILLOSCOPE CHAN A switch (S5) to position 9.

h. Verify that the Target Trigger pulse meets the frequency and amplitude requirements of paragraph 3.5.11.

4.7.3.12 Audio. Set up conditions as stated in 3.5.1.2, turn on 28V power supply, item 3 of Figure 3, and connect the oscilloscope, item 6 of Table VII, and the frequency counter, item 10 of Table VII to AUDIO terminal (J29). Verify that the signal at J29 is a 1 KHz ± 10 Hz square wave, with an amplitude of 0 V to 4.0 $^{+ 1.0}_{- 1.5}$ V.

4.7.3.13 Simulated target signals. The directions given below shall be followed in sequence.

a. Attach oscilloscope, item 6 of Table VII, to OSCILLOSCOPE CHAN A (J6), position 9.

b. Set up conditions as specified in 3.5.1.2.

c. Turn on -33V power supply, item 4 of Figure 3.

d. Turn on 28V power supply, item 3 of Figure 3.

e. Confirm presence of input signal, item 1 of Table II, at terminal J4-E.

f. Set MODE switch, S9 in 400M position as specified for RUN 1 of Table III.

g. Set DIRECTION switch, S12 as specified for RUN 1 of Table III.

h. Operate the TARGET/NO TARGET switch, S3 per the directions given in 3.5.13.4.

i. Operate the STOP switch, S11 per the directions given in 3.5.13.5.

j. When note 3 is specified in the description column of Table III, the conditions of 3.5.13.6 apply.

k. With conditions of RUN 1, Table III satisfied verify that the target pulse meets the amplitude and frequency requirements of 3.5.13.1, and the requirements given in the description column of Table III.

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1. In like manner, adjust the MODE switch, DIRECTION switch, and type of signal per Table III for each RUN in Table III, and verify that the output signal meets the requirements given in 3.5.13.1 and Table III.

4.7.3.14 Doppler signals.

4.7.3.14.1 Sum doppler. The directions given below shall be followed in sequence.

a. Connect the oscilloscope, item 6 of Table VII, across the 150 ohm load resistor connected to terminal F of P3 illustrated in Figure 1.

b. Adjust all controls and conditions per 3.5.13.1, RUN 2, as defined in 4.7.3.13.

c. Turn the SUM level control R2 fully clockwise.

d. Confirm the presence of input signal, Item 2 of Table II, at terminal P3-D, per Figure 1.

e. Determine the pulse conditions at the test points specified in 3.5.14.1e.

f. Verify that the Sum Doppler signal on the oscilloscope connected per a. meets the characteristics as defined in Table IV, Item 1.

4.7.3.14.2 Difference doppler (In Phase). The directions given below shall be followed in sequence.

a. Connect the oscilloscope, item 6 of Table VII across the 150 ohm load resistor, connected to terminal G of P3 illustrated in Figure 1.

b. Adjust all controls and conditions per 3.5.13.1, RUN 2, as defined in 4.7.3.13.

c. Turn the DIFFERENCE level control R3 full clockwise.

d. Confirm the presence of input signal, Item 2 of Table II, at terminal P3-D, per Figure 1.

e. Determine the pulse conditions at the test points specified in 3.5.14.2e.

f. Verify that the Difference Doppler signal on the oscilloscope connected per a. meets the characteristics as defined in Table IV, Item 2.

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4.7.3.14.3 Difference doppler (Out-of-Phase). The directions given below shall be followed in sequence.

a. Set conditions a. through d. of 4.7.3.14.2.

b. Determine the pulse conditions at the test points specified in 3.5.14.3e.

c. Verify that the Difference Doppler signal on the oscilloscope meets the characteristics as defined in Table IV, Item 3.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet 9333061.

6. NOTES

6.1 Intended use. The range computer tester is for use by a maintenance technician to locate defective units, assemblies, and subassemblies of Radar Set AN/VPS-2. The range computer test (Test Set, Subassembly MX-8049/TPM-22A) is a part of the Support Maintenance Shop Set. This equipment tests both the search and track modes of Range Computer CP-888/VPS-2, a part of the Radar Set AN/VPS-2.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the item detail specifications.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements, if other than specified in Section 4 of item detail specification.

f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificate of conformance for each lot or shipment of product.

6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714, tailored, with appropriate addendum to each. See 6.3.

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b. Data Cards - When required by incorporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government designs and contractor designs. Government designs are designated by drawing numbers in the appropriate paragraph in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government design is provided; however, unless otherwise specified, the contractor may submit alternate or modified contractor designs in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 Government designs. Government designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in DOD-STD-100.

c. An envelope drawing, as defined in DOD-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which Government designs are not specified and include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with Government designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number, paragraph number and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

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6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAF-I (D), Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as US Army Armament Research and Development Command, (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 4931-A201)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-50334B (AR)	Test Set, Subassembly MX-8049/TPM-22A (Range Computer Tester)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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